

Exercise Sheet 1

Univariate exploratory data analysis & distributions

Use of AI You may use AI tools to help draft code or wording, but your final answers should show your own judgment for your own learning.

Setup in Python Continue with the return series constructed in Sheet 0. If you work from a fresh notebook, use `numpy`, `pandas`, `matplotlib`, `scipy`, `statsmodels`, and `yfinance`. Use the two tickers SPY (S&P 500 ETF) and AAPL (Apple Inc.). Use the time window

`start_date = 2022-01-01,` `end_date = 2025-12-31.`

If the API returns no record on `end_date`, use the closest previous trading day.

Notation Let S_t denote the daily price on trading day t :

$$S_t := \begin{cases} \text{Adj Close at day } t, & \text{if available,} \\ \text{Close at day } t, & \text{otherwise.} \end{cases}$$

Do not mix `Close` and `Adj Close` within an asset. Use trading days only; do not fill non-trading days; drop missing prices before computing returns. Define daily log-returns by

$$r_t := \log(S_t) - \log(S_{t-1}).$$

Problem 1 For each asset:

- a) Construct density histograms (i.e., normalized so that the total area equals one) using $B \in \{30, 60, 120\}$ equally spaced bins. Use the empirical 0.1% and 99.9% quantiles to determine the displayed x-axis limits. For each value of B :
 - i) Plot the histogram.
 - ii) Describe what changes in the center and tails when B increases.
 - iii) Verify numerically that the sum of bar areas is one, up to numerical rounding.
- b) Let $\hat{f}_h$ denote the kernel density estimate (KDE) computed using the `scipy.stats` function `gaussian_kde`. By default, `gaussian_kde` uses Scott's rule, which selects the bandwidth factor $h_{\text{Scott}} = n^{-1/5}$ for univariate data. Treat this as a reference bandwidth and consider

$$h = c h_{\text{Scott}}, \quad c \in \{0.5, 1, 2\}.$$

Overlay the three KDE curves on the same axes and comment on the effect of changing the bandwidth.

- c) Choose one value of c that you consider reasonable and justify your choice. Your discussion should address:
- i) which features of the distribution you are trying to represent;
 - ii) the trade-off between the center and the tails.
- d) Prove the following results for an arbitrary fixed sample $r_1, \dots, r_n$. Let $h > 0$ and let K be a symmetric probability density, so $K(u) = K(-u)$, with finite second moment

$$m_2(K) := \int_{-\infty}^{\infty} u^2 K(u) du < \infty.$$

Write

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{t=1}^n K\left(\frac{x - r_t}{h}\right), \quad \bar{r} = \frac{1}{n} \sum_{t=1}^n r_t.$$

- i) Prove that the mean of the estimated density equals the sample mean:

$$\mu_h := \int_{-\infty}^{\infty} x \hat{f}_h(x) dx = \bar{r}.$$

- ii) Prove that the variance of the estimated density is

$$v_h := \int_{-\infty}^{\infty} (x - \mu_h)^2 \hat{f}_h(x) dx = \frac{1}{n} \sum_{t=1}^n (r_t - \bar{r})^2 + h^2 m_2(K).$$

- iii) Calculate $m_2(K)$ for the triangular kernel $K(u) = (1 - |u|)\mathbf{1}\{|u| \leq 1\}$ and for the Gaussian kernel $K(u) = \phi(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right)$, $u \in \mathbb{R}$. What happens to the mean and the additional variance due to smoothing when h is doubled? Does the total variance quadruple?

Hint: Substitute $u = (x - r_t)/h$, and use symmetry to show $\int_{-\infty}^{\infty} u K(u) du = 0$.

Problem 2 For each asset from Problem 1:

- a) Fit a univariate Gaussian distribution and a univariate Student- t distribution by maximum likelihood estimation and calculate the estimated parameters for both.
- b) Overlay both fitted densities on one of the density histograms from Problem 1a), using the empirical 0.1st–99.9th percentile range. Where do the fitted models differ most; which asset shows the stronger departure from Gaussianity?
- c) Let R denote a random variable following one of the fitted models. Compute $\mathbb{P}(R \leq q_{0.01}^{\text{emp}})$ and $\mathbb{P}(R \geq q_{0.99}^{\text{emp}})$, where q_{α}^{emp} denotes the empirical α -quantile of the observed returns. Compare these probabilities with the empirical values of 1% and discuss which fitted model better represents the tails.
- d) Verify the Gaussian MLE calculation from 2a) directly by computing the following:

$$\hat{\mu} = \bar{r}, \quad \hat{\sigma}_{\text{MLE}} = \sqrt{\frac{1}{n} \sum_{t=1}^n (r_t - \bar{r})^2}.$$

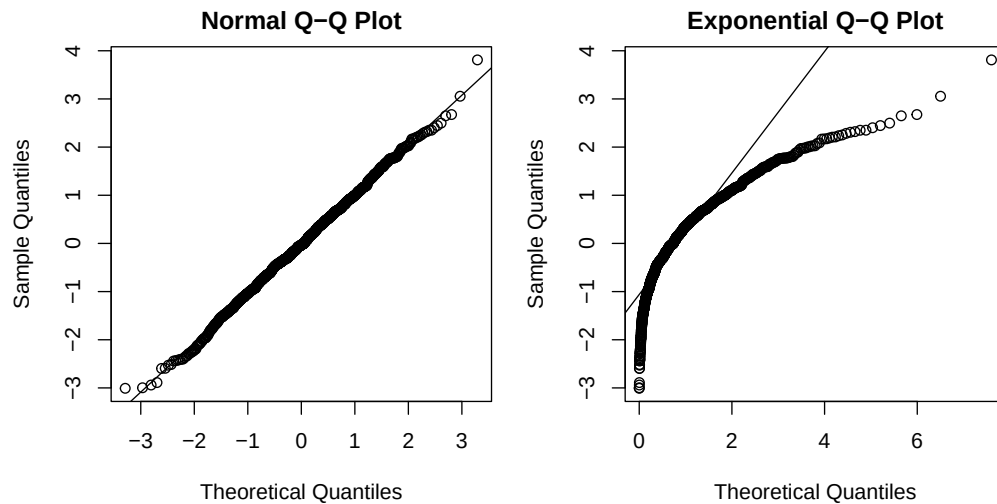
- e) Discuss what the following code tries to do, identify the problems in the code, and propose a corrected version.

```
def nll(theta, x):
    nu, loc, scale = theta
    return -np.sum(stats.t.logpdf(x, df=nu, loc=loc, scale=scale))

theta0 = np.array([2.0, 0.0, 0.0])
res = scipy.optimize.minimize(nll, theta0, args=(r,))
```

Problem 3 For each asset from Problem 1:

- Construct a Normal Q-Q plot and a Student- t Q-Q plot using the fitted models from Problem 2a). State what is shown on each axis.
- Compare the two Q-Q plots. Does the Student- t model reduce the tail deviations? Does either symmetric model explain visible asymmetry?
- The figure below shows two Q-Q plots for unknown datasets. For each plot, describe what features of the underlying distribution you can infer.



Problem 4 For each asset from Problem 1, define the one-day loss $L_t = -r_t$. Use confidence levels $\alpha \in \{0.95, 0.99\}$ for Value-at-Risk (VaR) and Expected Shortfall (ES) computations.

- Compute the empirical VaR and ES at both levels.
- Using the fitted Gaussian model from Problem 2a), compute the corresponding model-based VaR and ES.
- Compare the empirical and Gaussian values. Check that ES is at least as large as VaR at each level, and interpret any discrepancy between empirical and Gaussian tail risk.
- Use the fitted Student- t density and the Q-Q plots to predict qualitatively whether a Student- t risk estimate would be closer to the empirical value.